LECTURE-3

Water

INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

Key concepts

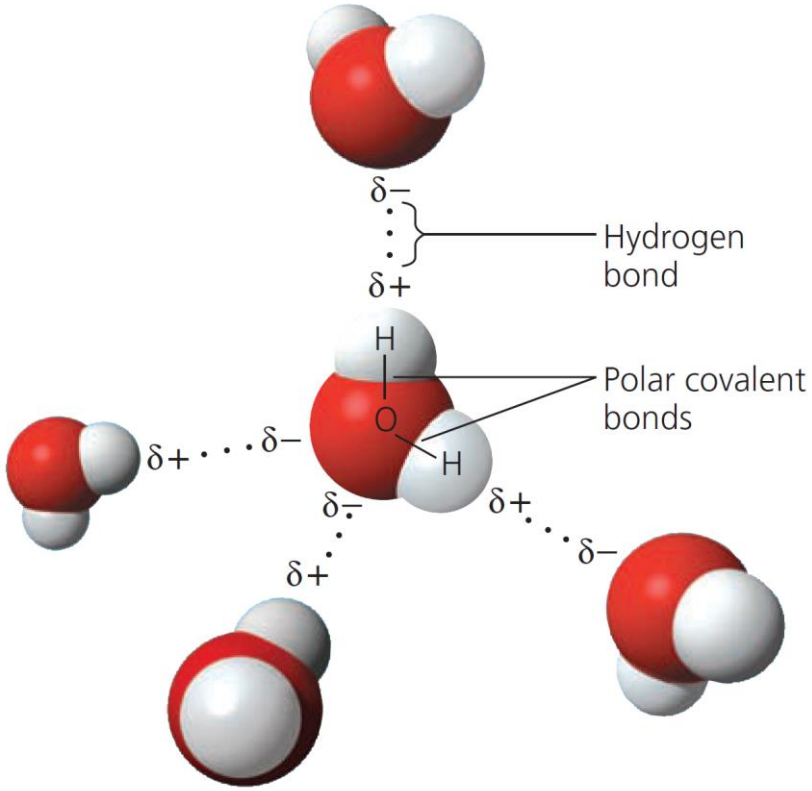
KEY CONCEPTS

3.1 Polar covalent bonds in water molecules result in hydrogen bonding

3.2 Four emergent properties of water contribute to Earth's suitability for life

3.3 Acidic and basic conditions affect living organisms

Hydrogen bonding and polar bonds



▲ **Figure 3.2 Hydrogen bonds between water molecules.** The charged regions in a water molecule are due to its polar covalent bonds. Oppositely charged regions of neighboring water molecules are attracted to each other, forming hydrogen bonds. Each molecule can hydrogen-bond to multiple partners, and these associations are constantly changing.

Four emergent properties of water contribute to Earth's suitability for life

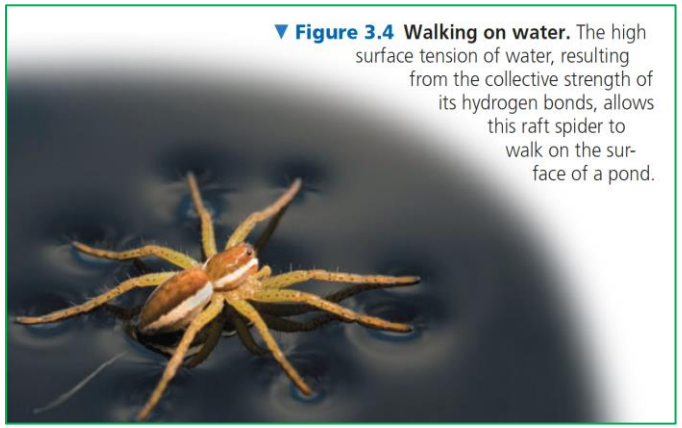
- ✓ Cohesive behavior.
- ✓ Ability to moderate temperature.
- ✓ Expansion upon freezing.
- ✓ Versatility as a solvent.

Cohesion, Adhesion and Surface tension

- **Cohesion**: Collectively, hydrogen bonds hold water together.

- *Cohesion among water molecules plays a key role in transporting water and dissolved nutrients against gravity in plants.*

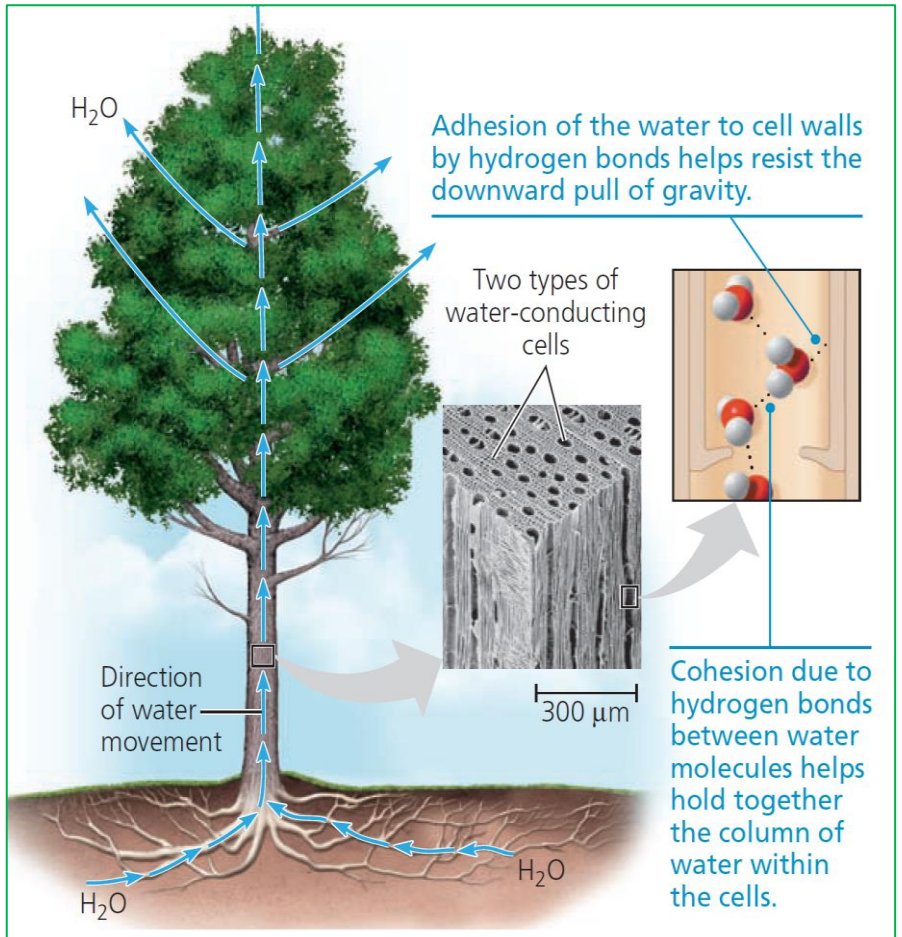
  - ✓ **Adhesion**, the clinging of one substance to another, also contributes, as water adheres to the walls of the vessels.



▼ **Figure 3.4 Walking on water.** The high surface tension of water, resulting from the collective strength of its hydrogen bonds, allows this raft spider to walk on the surface of a pond.

- **Surface tension**, a measure of the force necessary to stretch or break the surface of a liquid, is related to cohesion.

  - ✓ *Water has a greater surface tension than most other liquids because hydrogen bonds among surface water molecules resist stretching or breaking the surface.*
  - ✓ *Water behaves as if covered by an invisible film.*
  - ✓ *Some animals can stand, walk, or run on water without breaking the surface.*



▲ **Figure 3.3 Water transport in plants.** Evaporation from leaves pulls water upward from the roots through water-conducting cells. Because of the properties of cohesion and adhesion, the tallest trees can transport water more than 100 m upward—approximately one-quarter the height of the Empire State Building in New York City.

Thermal energy and Heat

- Thermal energy is the energy associated with the random movement of atoms and molecules.
- Whenever two objects of different temperature are brought together, thermal energy passes from the warmer to the cooler object until the two are the same temperature.
- Heat must be absorbed to break hydrogen bonds, and heat is released when hydrogen bonds form.
- To raise the temperature of water, hydrogen bonds between water molecules must be broken before the molecules can move faster.
- Thus, water absorbs a large amount of heat (much of it used to disrupt hydrogen bonds) while warming up only a few degrees.
- Conversely, when water cools, water molecules slow down and more hydrogen bonds form, releasing a considerable amount of heat.

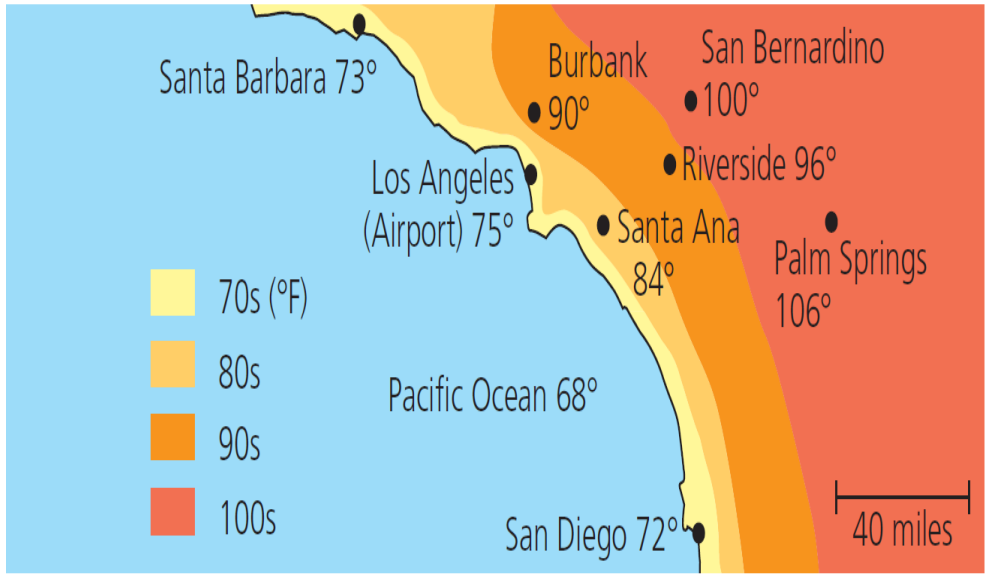
Specific heat: Water has high specific heat

- The specific heat of a substance is defined as the amount of heat that must be absorbed or lost for 1 g of that substance to change its temperature by 1 cal/g · °C.

- Compared with most other substances, water has an unusually high specific heat.

- A large body of water can absorb a large amount of heat from the sun during the daytime in the summer and yet warm only a few degrees.

- At night and during the winter, the warm water heats the cooler air.



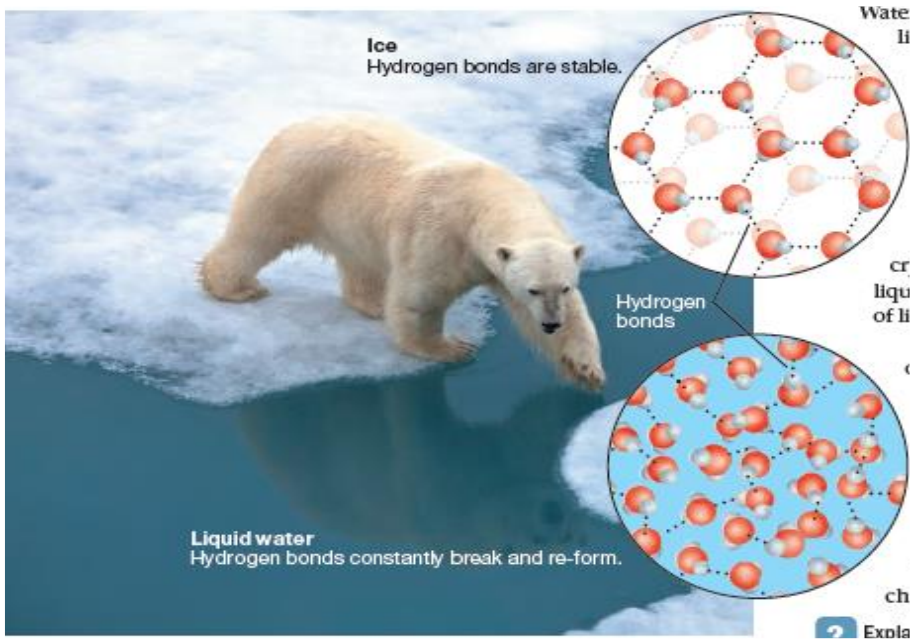
▲ **Figure 3.5** Temperatures for the Pacific Ocean and Southern California on an August day.

- Therefore, the **oceans and coastal land areas have more stable temperatures than inland areas**.

- **Living things** are made of primarily water, so they **resist changes in temperature better** that they would if composed of a liquid with a lower specific heat.

Water expands by Freezing

- **Water is less dense as a solid than as a cold liquid**.
- Water begins to freeze when its molecules are no longer moving vigorously enough to break their hydrogen bonds.
- When water reaches 0°C, it becomes locked into a crystalline lattice, with each water molecule bonded to four partners.
- As ice starts to melt, some of the hydrogen bonds break, and water molecules can slip closer together than they can while in the ice state.

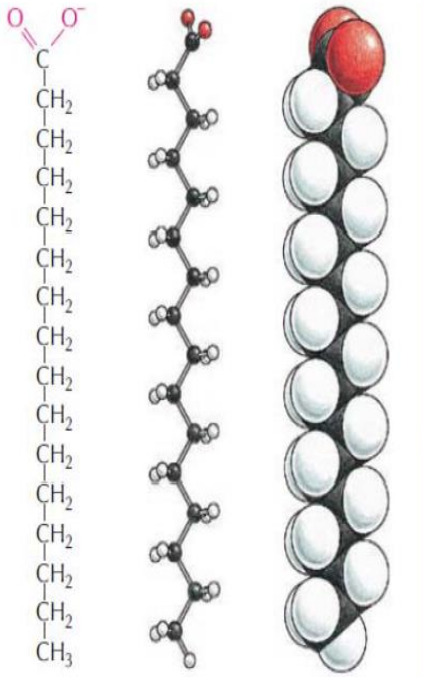


Hydrophilicity and hydrophobicity

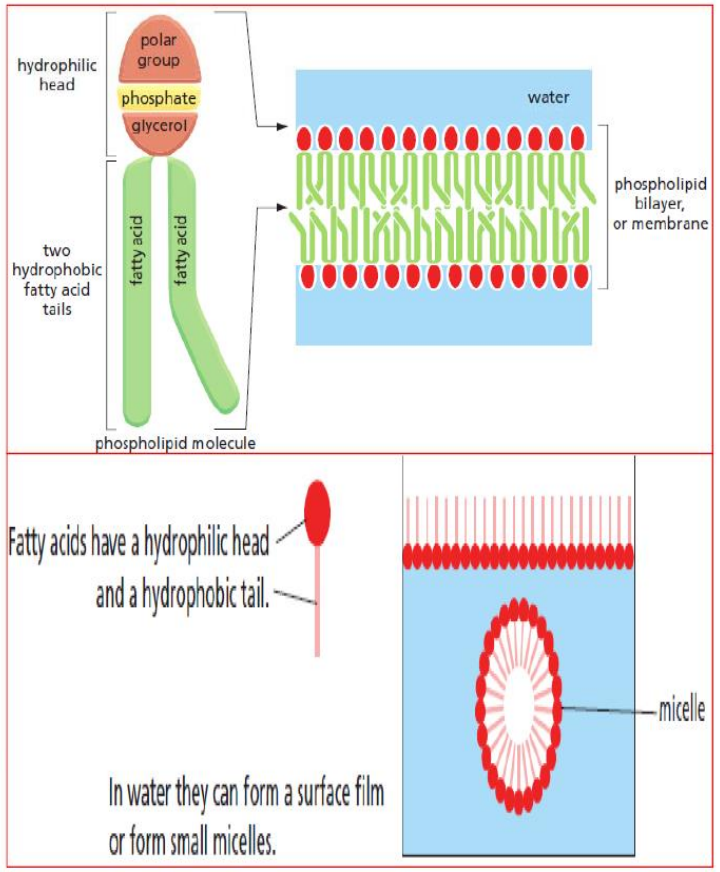
- A solution is a liquid consisting of a uniform mixture of two or more substances.
- The dissolving agent (in our example, water) is the solvent, and a substance that is dissolved (in this case, salt) is a solute.
- An aqueous solution (from the Latin aqua, water) is one in which water is the solvent.
- Water is very versatile because of the polarity of water molecules. **Polar molecules are soluble in water** because they form hydrogen bonds with water.
- Even large molecules, like proteins, can dissolve in water if they have ionic and polar regions.

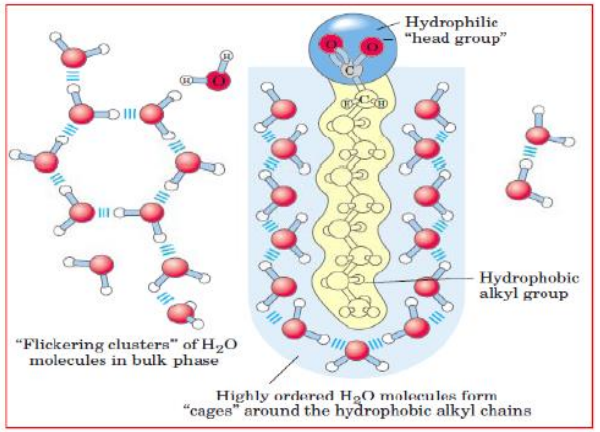
- A substance that has an affinity for water is **hydrophilic** *(water-loving).*
  - Hydrophilic substances are dominated by ionic or polar bonds.
- Substances that have no affinity for water are **hydrophobic** *(water-fearing).*
  - ✓ Hydrophobic substances are nonionic and have nonpolar covalent bonds.
  - ✓ Oils such as vegetable oil are hydrophobic because the dominant bonds, carbon-carbon and carbon-hydrogen, share electrons equally.
  - ✓ Hydrophobic molecules are major ingredients of cell membranes.

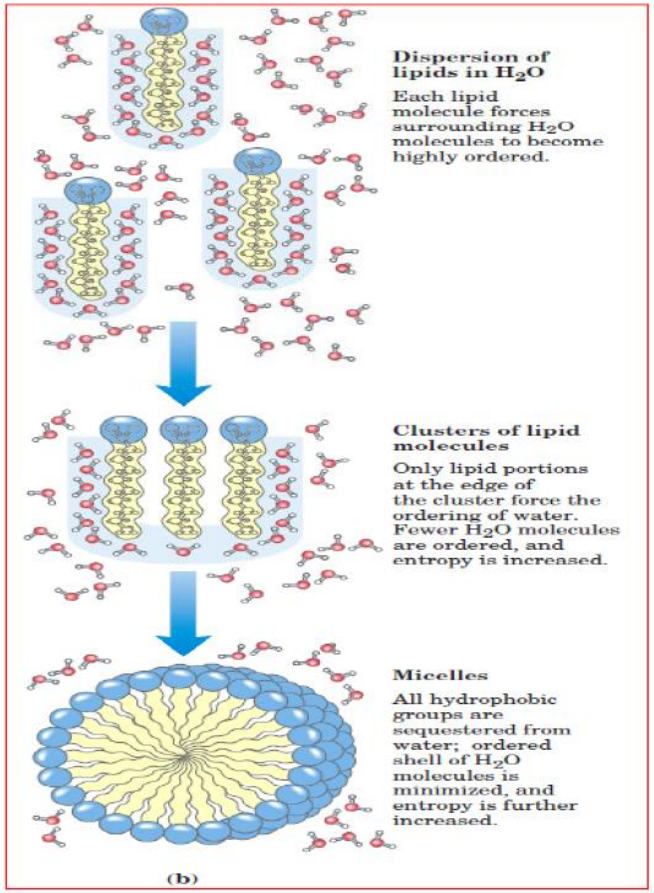
Hydrophobic interactions: increase in water entropy



Structural formula | Ball and Stick model | Space filling model





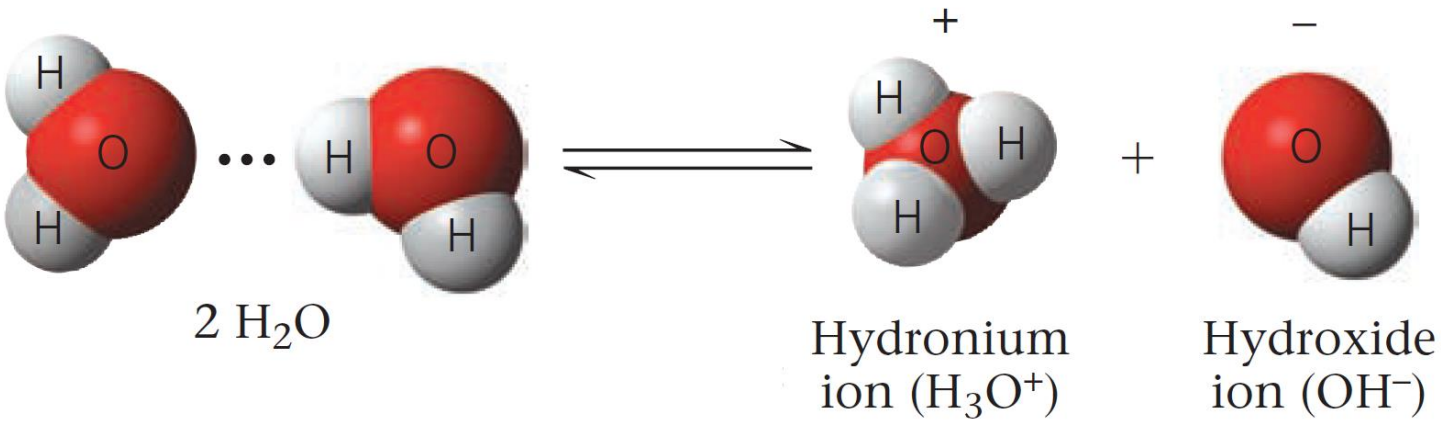


- ❖ Amphipathic compounds contain regions that are polar (or charged) and regions that are nonpolar.
- ❖ The strength of hydrophobic interactions is not due to any intrinsic attraction between nonpolar moieties.
- ❖ Rather, it results from the system's achieving greatest thermodynamic stability by minimizing the number of ordered water molecules required to surround hydrophobic portions of the solute molecules.

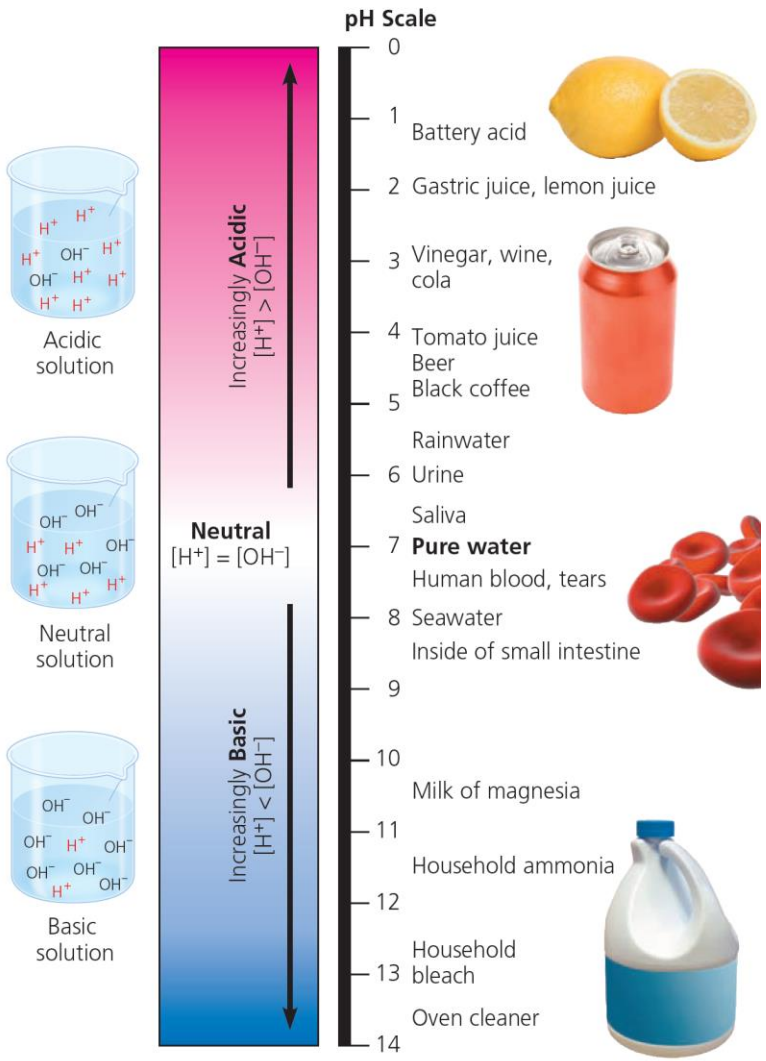
Acidic and basic conditions

- Occasionally, a hydrogen atom participating in a hydrogen bond between two water molecules shifts from one molecule to the other.
  - ✓ The hydrogen atom leaves its electron behind and is transferred as a single proton—a **hydrogen ion** (H^+).
  - ✓ The water molecule that lost the proton is now a **hydroxide ion** (OH^-).
  - ✓ The water molecule with the extra proton is now a **hydronium ion** (H_3O^+).

- Although the dissociation of water is reversible and statistically rare, it is very important in the chemistry of life.



pH



▲ **Figure 3.10** The pH scale and pH values of some aqueous solutions.

- The **pH scale** is used to describe how acidic or basic a solution is.
- An **acid** is a substance that increases the hydrogen ion concentration in a solution.
  - ✓ When hydrochloric acid is added to water, hydrogen ions dissociate from chloride ions: $HCl \rightarrow H^+ + Cl^-$.(acidic solution)
- Any substance that reduces the hydrogen ion concentration in a solution is a **base**.
- The pH scale measures the H+ concentration of a solution.
- The **pH of a solution** is defined as **the negative logarithm (base 10) of the hydrogen ion concentration**.
- $pH = -\log [H^+]$ or $[H^+] = 10^{-pH}$
- Neutral solution: $[H^+] = 10^{-7}$ *M* and pH = 7.

Buffer

- **A buffer** is a substance that minimizes changes in the concentrations of H+ and OH- in a solution.
- It does so by accepting hydrogen ions from the solution when they are in excess and donating them to the solution when depleted.
- Most buffer solutions contain a weak acid and its corresponding base, which combine reversibly with hydrogen ions.
- Several buffers contribute to pH stability in human blood and many other biological solutions. One of these is carbonic acid (H_2CO_3) forms when CO_2 reacts with water in blood plasma.
- Carbonic acid dissociates to yield a bicarbonate ion (HCO^{3-}) and a hydrogen ion (H^+):

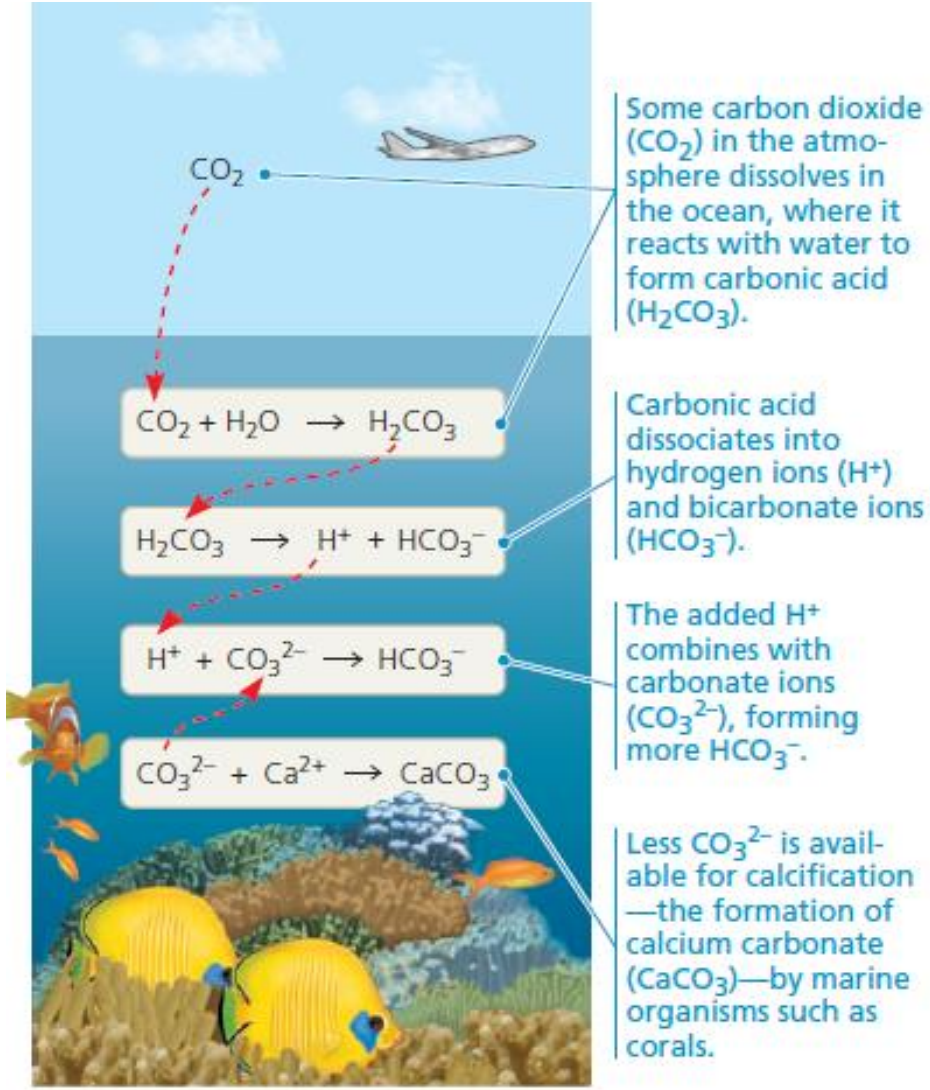
$$H_2CO_3 \underset{\text{Response to a drop in pH}}{\overset{\text{Response to a rise in pH}}{\rightleftharpoons}} HCO_3^- + H^+$$

H_2CO_3	HCO_3^-	H^+
H^+ donor (acid)	H^+ acceptor (base)	Hydrogen ion
Carbonic acid	bicarbonate ion	

The chemical equilibrium between carbonic acid and bicarbonate acts as a pH regulator, the reaction shifting left or right as other processes in the solution add or remove hydrogen ions.

Acidification: A Threat to Water Quality

- Among the many threats to water quality posed by human activities is burning fossil fuels, which release gaseous compounds into the atmosphere.
- Carbon dioxide is the main product of fossil fuel combustion. The oceans absorb about 25% of human-generated CO2.
- Despite the huge volume of water in the oceans, scientists worry that the absorption of so much CO2 will harm marine ecosystems.
- When CO2 dissolves in seawater, it reacts with water to form carbonic acid, which lowers ocean pH, a process known as **ocean acidification.**
- scientists calculate that the pH of the oceans is 0.1 pH unit lower now than at any time in the past 420,000 years.
- Recent studies predict it will drop another 0.3–0.5 pH unit by the end of this century.



Derivations

Equilibrium, pH, and buffer

pH in terms of equilibrium constant

- Many biochemical reactions in nature is reversible reactions. They do not proceed to completion.
  There is a balance of forward and backward reaction, and the net velocity is zero. It reaches a equilibrium state. (What is a steady state ?)

- The reaction $HA \leftrightarrows H^+ + A^-$.
  $v_f \ \alpha \ [HA]$, and $v_f = k_1 \ [HA]$ (Forward velocity), and k_1 is proportionality constant.
  $v_r \ \alpha \ [H^+]$ and $v_r \ \alpha \ [A^-]$ and therefore, $v_r \ \alpha \ [H^+][A^-]$ or $v_r = k_{-1}[H^+][A^-]$.

- At equilibrium $v_f = v_r$, and $\frac{k_1}{k_{-1}} = \frac{[H^+][A^-]}{[HA]}$ Therefore,

$$\boxed{k_{eq} = \frac{[H^+][A^-]}{[HA]}}$$

Water and Kw

- Reactions of water– Ionization: Reversible self-dissociation. Water generates H^+ and OH^-.

- The equilibrium reaction of water is

$$\boxed{H_2O \rightleftarrows H^+ + OH^-}$$

.

- Hydrogen ion do not exist as free H+ ion in the solution .

- Actually exists as H_3O^+, and for simplicity we just write H^+.

- Express extent of ionization quantitatively– Use law of mass action to define the equilibrium point of the dissociation reaction.

Water and Kw

- K_{eq} is defined as the ratio of the products and reactant .Units used to define concentration are Molarity (M) = moles/L.

- K_{eq} for pure water is determined experimentally – 1.8 X 10^{-16} M at 25^0C. Concentration for pure water = 55.5 M (Weight in 1L/ M.W of Water(18)).

- The equation is $[H^+][OH^-] = K_{eq}\ [H_2O]$
  and $[H^+][OH^-] = (1.8 \text{ X } 10^{-16})(55)$
  and $[H^+][OH^-] = (1.0 \text{ X } 10^{-14})\ (M^2)$

- Chemical equation tells that $[H^+]$ and $[OH^-]$ dissociates equally, they must have equal concentrations in pure water;
  i.e., $[H^+] = [OH^-] = \sqrt{(1.0X10^{-14})} = 1.0X10^{-7}$ M.

pH scale

- The basis of the pH scale is that these numbers are very small and difficult to work.
- Sorensen introduced the term pH to conveniently express $[H^+]$. He defined pH as the negative logarithm of the hydrogen ion concentration.
- pH = -log$[H^+]$. Also the same as $\log(\frac{1}{[H^+]})$. p is an operator, and it means to take the negative log of.
- Example: pOH = -log$[OH^-]$. pH of the pure water is 7.
- pH + pOH = 14. pH is log based, and it ranges from 0 and 14.

Weak acid and bases

- Besides water, for other molecules their acid and base nature can be determined.
- Definition – According to Bronsted: Acid is the release of proton in water. Base is the acceptance of proton in water.
- Water can act as an acid and a base; i.e., amphiprotic. Strength of the acid is defined as its tendency to release a proton (dissociate).
- Acids can be strong or weak eg: HCl is a strong acid , CH_3COOH is a weak acid.
- Dissociation for an acid as: HA (**Acid**) $\leftrightarrows$ $\mathbf{H^+ + A^-}$ **(conjugate base)**

Conjugate Base

- Conjugate Base: Base formed by the removal of proton from an acid . The acid and conjugate base are complementary species every acid has a conjugate base.

- Should be able to identify acids and their conjugate base given their structures.

- Acetic acid – Acetate buffer: $CH_3COOH \leftrightarrows H^+ + CH_3COO^-$ (Conjugate base).

- Dissociation constant of acid $K_a = \frac{[H^+][A^-]}{[HA]}$.

- Apply p rule and take the negative log of K_a to get pKa (= -log K_a).

Relationship between K_a and K_b

- Consider a weak acid HA dissolved in water and the ionization of acid in water is given by $HA + H_20 \leftrightarrows H_30 + A^-$, and $K_a = \frac{[H_3O][A^-]}{[HA]}$.
- On the other hand if we start dissociating the conjugate base A- in water, we get $H_30 + A^- \leftrightarrows HA + OH^-$ and and $K_b = \frac{[HA][OH^-]}{[A^-]}$.
- Solving the K_a and K_b expressions for H_3O^+ and OH^-. $[H_3O^+] = \frac{[HA][k_a]}{[A^-]}$ and $[OH^-] = \frac{[A^-]k_b}{[HA]}$.
- Substituting for $K_w = [H_3O^+][OH^-]$, we get,

$$\boxed{K_a \times K_b = K_w.}$$

- Taking logarithms, we get $\log(K_a) + \log(K_b) = \log(K_w)$.